Sylvain Gaulhiac

Postdoctoral Fellow in University of Alberta

Positions

2021–Present **Postdoctoral Fellowship**, *University of Alberta*, Edmonton, Canada.

2019–2020 **ATER (Attaché Temporaire d'Enseignement et de Recherche)**, *Sorbonne Université*, Paris.

Education

2016–2020 **Ph.D under the supervision of Antoine Ducros**, *Sorbonne Université*, Paris.
Title : Comportements anabéliens des courbes de Berkovich sous les perspectives tempérées

2015 **Agrégation Externe de Mathématiques**, Rank : 15.

2012–2016 **École Normale Supérieure de Rennes and Université de Rennes 1**.
- 2014-2015 : Master 2 in Algebra and Geometry (Mention Bien)
- 2014-2015 : Preparation to the Agrégation Externe de Mathématiques, Master 2 Mathematics and Teaching (Mention Très bien)
- 2013-2014 : Master 1 in Mathematics
- 2012-2013 : Licence 3 in Mathematics (Mention Très bien)

2012 **Accepted in École Normale Supérieure de Rennes (ENS)**.

2009-2012 **Classes Préparatoires MPSI/MP***, *Lycée Joffre*, Montpellier, France.

Mathematical interests

- Algebraic/Arithmetic geometry, Number theory
- Anabelian geometry
- Analytic non-archimedean geometry (Berkovich/adic geometry)
- All kinds of fundamental groups, with an emphasis on the tempered fundamental group
- Condensed mathematics

Research papers

Reconstruction anabélienne du squelette des courbes analytiques, accepted in Annales de l'Institut Fourier, arXiv:1904.03126

University of Alberta – Edmonton, Canada
✉ *gaulhiac@ualberta.ca* • *Office : CAB 586*

Towards tempered anabelian behaviour of Berkovich annuli, submitted to Bulletin de la Société Mathématique de France, arXiv:2101.06648

Comparison between admissible and de Jong coverings of rigid analytic spaces in mixed characteristic, arXiv:2201.08065

Research talks

January 2022 **Hausel Working Seminar**, *IST Austria*, Klosterneuburg.
Title : Tempered anabelian geometry in analytic Berkovich spaces.

October 2021 **Arithmetic Geometry Seminar**, *University of Alberta*, Edmonton.
Title : Do Berkovich curves have anabelian behaviours?

March 2020 **Number Theory Seminar**, *Sorbonne Université*, Paris.
Title : Les courbes de Berkovich ont-elles un comportement anabélien ?

October 2019 **Number Theory Seminar**, *Université de Caen Normandie*, Caen.
Title : Anabelian Geometry on Berkovich curves.

June 2019 **GAeL XXVII, Géométrie Algébrique en Liberté**, *IMAR*, Bucharest.
Title : Anabelian behaviours of Berkovich curves.

April 2019 **Seminar Ph.D-Master Students**, *Sorbonne Université*, Paris.
Title : Anabelian Geometry : from hyperbolic curves to Berkovich curves.

March 2019 **Ph.D Seminar**, *Université de Rennes 1*, Rennes.
Title : Anabelian geometry in Berkovich spaces.

April 2018 **Ph.D Seminar**, *Université de Caen Normandie*, Caen.
Title : Berkovich spaces, or how to define analytic spaces over non-archimedean fields.

June 2017 **Seminar**, *Chebyshev Institute*, Saint-Petersburg.
Titlte : Introduction to Berkovich Spaces.

Predoctoral Research experience

2015 **Master 2 thesis under the supervision of Pierre Lochak**, *Université Pierre et Marie Curie*, Paris.
Title : Autour de la connexion KZB universelle

Summer 2014 **Master 1 thesis under the supervision of Zinovy Reichstein**, *University of British Columbia*, Vancouver.
Title : Simplifying polynomials in one variable using quadratic form theory (Hermite-Joubert problem in quadratically closed field extensions)

Spring 2013 **L3 thesis under the supervision of Mathieu Florence**, *Université Pierre et Marie Curie*, Paris.
Title : Autour du problème de Noether pour les groupes alternés

Organisation of Seminars

2021 Co-organisation with Adam Topaz of the **Condensed Mathematics Seminar**, *University of Alberta*, Edmonton.

2018–2019 In charge of the monthly seminar **RÉGA (Réseau d'Etudiants en Géométrie Algébrique)**, *Institut Henri Poincaré*, Paris.

Teaching experience

2021 **Calculus 1, Math100**, Instructor, *University of Alberta*

2019-2020 **Differential equations (level L2)**, Teaching Assistant, *Sorbonne Université*

2019-2020 **Algebra (level L3)**, Teaching Assistant, *Sorbonne Université*

2019-2020 **Arithmetic and algebra (level L2)**, Teaching Assistant, *Sorbonne Université*

2017-2018 **Euclidian and Hermitian vector spaces, affine isometries (level L2)**, Teaching Assistant, *Sorbonne Université*

2017-2018 **Isometry groups and symmetrical groups (level L2)**, Teaching Assistant, *Sorbonne Université*

2017-2018 **Arithmetic and algebra (level L2)**, Teaching Assistant, *Sorbonne Université*

2016-2017 **Linear algebra 2, Euclidian vector spaces, affine isometries (level L2)**, Teaching Assistant, *Sorbonne Université*

2016-2017 **Analysis and algebra for science (level L1)**, Teaching Assistant, *Sorbonne Université*

2018–2020 Courses given to students preparing the french **Agrégation Externe de Mathématiques** (highest teaching competitive exam).

2015–2016 Interrogations (khôlles) given in preparatory classes MPSI in Lycée Louis-le-Grand (Paris).

Attended conferences and workshops

November 2021 **BIRS Alberta Number Theory Days XIII**, *BIRS*, Banff, (online).

November 2019 **Géométrie Algébrique et Géométrie Complexe (GAGC)**, *CIRM*, Marseille.

June 2019 **GAeL XXVII, Géométrie Algébrique en Liberté**, *IMAR*, Bucharest, Romania.

May 2019 **School : Curves and Groups in Families**, *Université Rennes 1*, Rennes.

December 2018 **Géométrie Algébrique et Géométrie Complexe (GAGC)**, *CIRM*, Marseille.

September 2018 **CATS 5, Categories and Stacks in Geometry, Topology and Algebra**, Lisbon, Portugal.

August 2018 **International Congress of Mathematics**, Rio de Janeiro, Brazil.

July 2018 **Berkovich Spaces 30 years**, *IHP*, Paris.

April 2018 **Modern and Classical Aspects on Algebraic Surfaces**, *Pedagogical University of Krakow*, Poland.

April 2018 **Crystals and Geometru in characteristic** p, *TU Munich*, Germany.

September 2017 **Instruments of Algebraic Geometry**, *IMAR*, Bucharest, Romania.

July 2017 **Berkovich Spaces, Tropical Geometry and Model Theory**, *Universidad de Los Andes*, Bogota, Colombia.

March 2017 **Géométrie analytique et équations différentielles p-adiques**, *CIRM*, Marseille.
March 2017 **Arizona Winter School: Perfectoid Spaces**, *University of Arizona*, Tuscon.

Languages

French	Mothertongue
English	Fluent
Spanish	Intermediate
Russian	Moderate *(CLES 1 obtained in 2017)*

Interests

Music

Out of mathematics I am a pianist as well as a musical critic in classical music for Bachtrack, specialized in piano and chamber music, but also covering orchestra. I wrote numerous reviews of concerts, as well as some interviews of musicians or program of concerts.

Other

- Litterature
- Photography
- Hiking
- Cooking
- Travelling